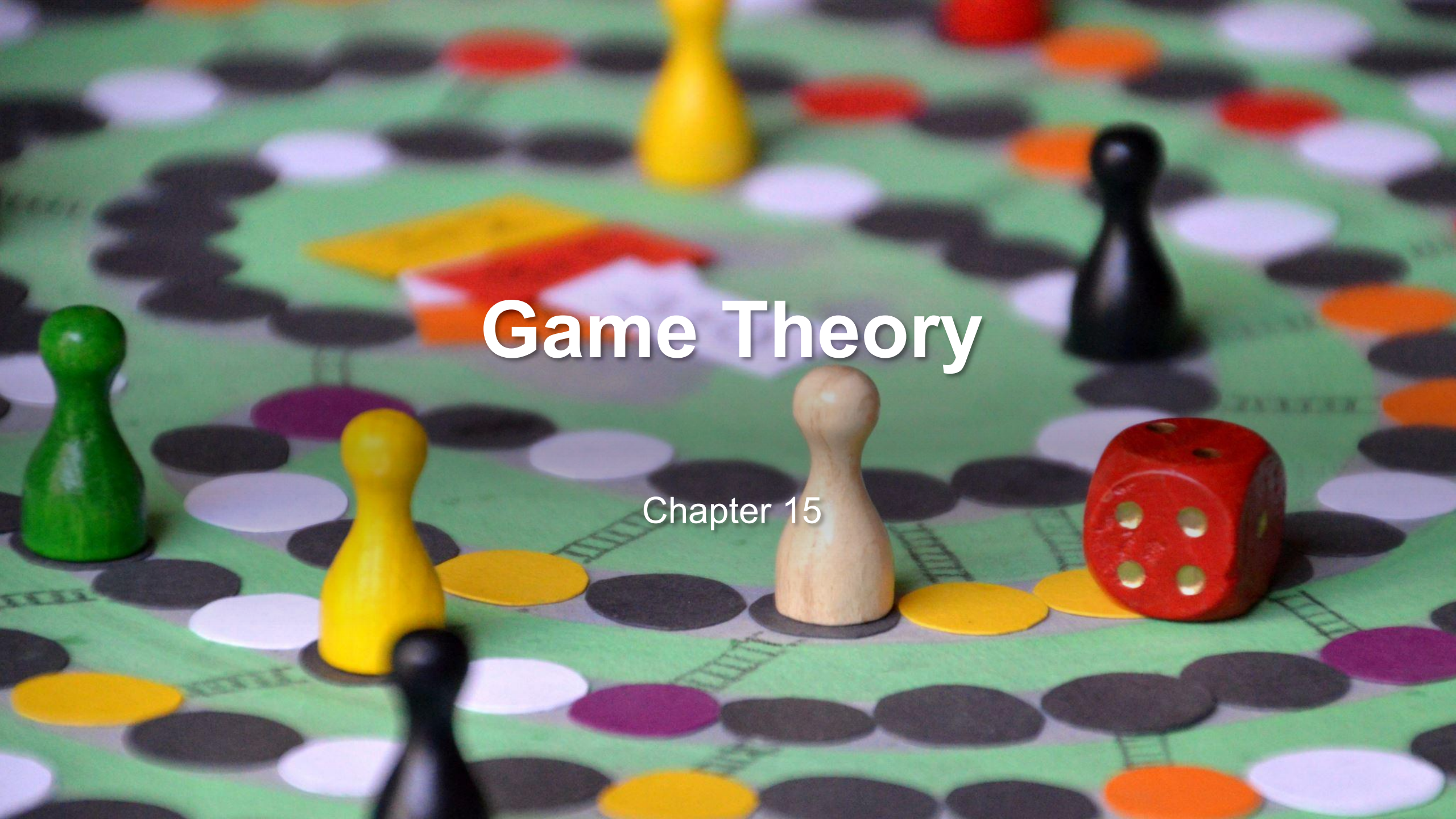


A close-up photograph of a board game. The board is green with a pattern of dark grey and light grey circles. Various game pieces are scattered across the board: a green pawn, a yellow pawn, a black pawn, a wooden pawn, and a red die showing the number six. The text "Game Theory" is overlaid in the center in a large, white, sans-serif font.

Game Theory

Chapter 15

Game Theory

- Life is full of conflict and competition
 - Military battles
 - Political campaigns
 - Marketing and advertisements
- The final outcome depends primarily upon the combination of strategies selected by the adversaries
- Game theory is a mathematical theory that deals with the general features of these kind of competitive situation
- It helps in decision making processes.

Game Theory

- **Competitive situation** are present every where, therefore game theory has applications in a **variety of areas including business and economics**.
- The 1994 John F. Nash, Jr., John C. Harsanyi, and Reinhard Selton won Nobel Prize for Economic Sciences for their analysis of equilibria in the theory of non-cooperative games.
- Then Robert J. Aumann and Thomas C. Schelling won the 2005 Nobel Prize for Economic Sciences for enhancing our understanding of conflict and cooperation through game-theory analysis.

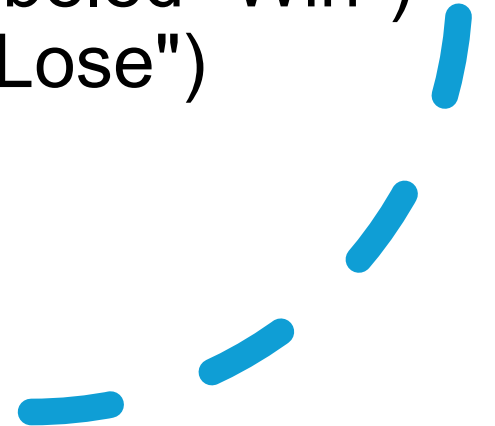


The Focus of This Chapter

- The focus in this chapter is on the simplest case, called **two-person, zero-sum games**.
- As the name implies, these games involve only two adversaries or players (who may be armies, teams, firms, and so on).
- They are called zero-sum games because one player wins whatever the other one loses, so that the sum of their net winnings is zero.

What are Two- Person, Zero-Sum Games?

- Games with only 2 players (e.g., you vs. me)
- Zero-sum: One player's gain equals the other's loss (total = 0)
- Focus on choices (strategies) each player makes
- If I win \$5, you lose \$5 – simple but tricky!
- A seesaw with one side up (labeled "Win") and the other down (labeled "Lose")



Example in Real Life

Cards, military battles,
elections

Business fights (e.g., Coke
vs. Pepsi ads)

Outcome depends on both
players' plans

Everyday competitions
where strategy matters!

The Odds and Evens Game

Each player shows 1 or 2
fingers at the same time

Even total (e.g., $2+2=4$)?
Player 1 (Evens) wins \$1

Odd total (e.g., $1+2=3$)?
Player 2 (Odds) wins \$1

A fun, quick way to test
game theory

Strategies of Odds and Evens

- Player 1's strategies: Show 1 finger or 2 fingers
- Player 2's strategies: Show 1 finger or 2 fingers
- 2 choices each—simple but strategic
- Your move decides who wins—guess or plan?
- Shows what Player 1 gets (win \$1 or lose \$1)
- Player 2 gets the opposite (zero-sum)

■ **TABLE 15.1** Payoff table for the odds and evens game

		Player 2	
		1	2
Strategy	1	1	-1
	2	-1	1

How this Game is Played

Both players pick a strategy secretly

Reveal choices at the same time

Payoff table shows who wins what

No sneak a quick look—your choice vs. theirs decides it!

Simple vs. Complex Strategies

Simple: One action (e.g., show 1 finger)

Complex: Plans for many moves (e.g., chess moves for every board)

Odds and evens is simple – 2 choices only

Games can get big, but we start small!

Payoff Table Details

Shows Player 1's gains (positive or negative)

Player 2's table is just the opposite (e.g., -1 for Player 1 = +1 for Player 2)

Numbers represent "value" (e.g., money, votes)

One table works for both in zero-sum games

Goals for the Game Theory

Find smart ways to pick strategies

Assumes players are rational
(think logically)

Players aim to win the most
for themselves

No helping the opponent—
just your best move!

Game Theory vs. Decision Analysis

Game theory: Two smart players compete

Decision analysis: One player vs. random nature

Odds and evens needs strategy, not luck

15.2 Solving Simple Games – A Prototype Example

Section 15.2

- This section introduces a **prototype example** (political campaign) to show how two-person, zero-sum games are set up and solved.
- Example: Two politicians plan campaign days, with votes as the payoff. **Three variations test different solving methods:**
 - Variation 1: Uses dominated strategies (eliminate weak options).
 - Variation 2: Uses minimax criterion (pick the safest best outcome).
 - Variation 3: Shows when simple methods fail, needing mixed strategies.
- Focus: Payoff tables, strategy choices, and rational decision-making.

Solving a Simple Game

How Politicians Play to Win Votes?

Campaign Setup

- Two politicians compete for a Senate seat in USA
- They plan 2 campaign days in Bigtown or Megalopolis
- Choices must be locked in advance—no spying allowed
- Each wants to outsmart the other to win more votes

Strategies for Each Politician

Three options (strategies) for each:

- Strategy 1: 1 day in each city
- Strategy 2: 2 days in Bigtown
- Strategy 3: 2 days in Megalopolis

They pick secretly at the same time

The main point is: Where they go affects vote gains or losses

Payoff Table Basics

Variation 1: Uses dominated strategies (eliminate weak options).

■ TABLE 15.3 Payoff table for player 1 for variation 1 of the political campaign problem

		Player 2		
		1	2	3
Player 1	Strategy 1	1	2	4
	2	1	0	5
	3	0	1	-1

Table shows votes (in thousands) Politician 1 wins

Politician 2 loses whatever Politician 1 gains (zero-sum)

The goal for each politician: Pick the best strategy to win votes

Numbers tell us who wins based on their choices

Variation 1 – Payoff Table (Dominated Strategies)

■ TABLE 15.3 Payoff table for player 1 for variation 1 of the political campaign problem

		Player 2		
		1	2	3
Player 1	1	1	2	4
	2	1	0	5
	3	0	1	-1
	Strategy			

- If both pick strategy 1, Politician 1 wins 1,000 votes
- Positive = win and negative = lose for politician 1
- Strategy 3 for politician 1 is weak (0,1, -1) versus strategy 1 (1,2,4)
- Politician 2 drop strategy 3 (4, 5, worse than 1, 0)
- Both end up with strategy 1 (Politician 1 wins 1,000 votes)

Breaking Down Each Row and Column

Politician 1's Strategy 1 (1 day each city)

- vs. Player 2 Strategy 1: Wins 1,000 votes
- vs. Player 2 Strategy 2: Wins 2,000 votes
- vs. Player 2 Strategy 3: Wins 4,000 votes

Politician 1's Strategy 2 (2 days Bigtown):

- vs. Player 2 Strategy 1: Wins 1,000 votes
- vs. Player 2 Strategy 2: No change (0 votes)
- Vs. Player 2 Strategy 3: Wins 5,000 votes

Politician 1's Strategy 3 (2 days Megalopolis):

- vs. Player 2 Strategy 1: No change (0 votes)
- vs. Player 2 Strategy 2: Wins 1,000 votes
- vs. Player 2 Strategy 3: Loses 1,000 votes

Solving with Dominated Strategies

What's a Dominated Strategy?

- A strategy is “dominated” if another strategy is always better (or at least as good) no matter what the opponent does.

We drop dominated strategies because a smart player wouldn't pick them.

Step 1: Check Politician 1's Strategies

- Compare Strategy 3 (0, 1, -1) with Strategy 1 (1, 2, 4)
- Strategy 1 beats Strategy 3 every time—Strategy 3 is dominated, so drop it.

■ TABLE 15.3 Payoff table for player 1 for variation 1 of the political campaign problem

		Player 2		
		1	2	3
Player 1	1	1	2	4
	2	1	0	5
	3	0	1	-1

Step 2: Check Politician 2's Strategies

- Politician 2 wants Politician 1 to win less (smaller numbers are better for Politician 2)
- Compare Strategy 3 (4, 5) with Strategy 1 (1, 1) and Strategy 2 (2, 0):
 - vs. Player 1 Strategy 1: $4 > 1$ (Strategy 1 better) and $4 > 2$ (Strategy 2 better)
 - vs. Player 1 Strategy 2: $5 > 1$ (Strategy 1 better) and $5 > 0$ (Strategy 2 better)
- Strategy 3 is dominated by both 1 and 2—drop Strategy 3.

■ **TABLE 15.3** Payoff table for player 1 for variation 1 of the political campaign problem

		Player 2		
Strategy		1	2	3
Player 1	1	1	2	4
	2	1	0	5
	3	0	1	1

Step 3: Check Again for Politician 1

- Strategy 3 is dominated by both 1 and 2, so we have to drop Strategy 3.
- Compare Strategy 2 (1, 0) with Strategy 1 (1, 2):
 - Column 1: $1 = 1$ (equal)
 - Column 2: $0 < 2$ (Strategy 1 better)
- Strategy 2 is dominated by Strategy 1—drop Strategy 2.

■ **TABLE 15.3** Payoff table for player 1 for variation 1 of the political campaign problem

		Player 2		
		1	2	3
Player 1	1	1	2	4
	2	1	0	5
	3	0	1	1

Step 4: Check for Politician 2

- For Politician 1, Strategy 2 is dominated by Strategy 1—drop Strategy 2. (A new table with only one row)
- Compare Strategy 2 with Strategy 1 (1):
 - $2 > 1$ (Strategy 1 better for Politician 2)
- Strategy 2 is dominated—drop Strategy 2.
- Finally, Player 2 opt strategy 1.

■ **TABLE 15.3** Payoff table for player 1 for variation 1 of the political campaign problem

		Player 2		
		1	2	3
Strategy	1	1	2	4
	2	1	0	5
Player 1	3	0	1	1

Conclusion

- Both politicians pick Strategy 1 (1 day in each city).
- Politician 1 wins 1,000 votes from Politician 2.
- This is the value of the game – not fair (not 0), but stable because neither can do better by switching.
- **Why it works**
- By dropping weak strategies step-by-step, we narrow it down to one clear choice strategy.
- Both are rational, so they avoid bad moves and end up here.

Variation 2 (Minimax Criterion)

Variation 2

- Unlike Variation 1, this version **doesn't have dominated strategies**, so we use a different method called the **minimax criterion** to solve it.

■ TABLE 15.4 Payoff table for player 1 for variation 2 of the political campaign problem

		Player 2			Minimum
		1	2	3	
Player 1	Strategy 1	-3	-2	6	-3
	2	2	0	2	0 ← Maximin value
	3	5	-2	-4	-4
Maximum:		5	0	6	
			↑		
			Minimax value		

The Setup Recap

- Two politicians are campaigning for a Senate seat.
- Each has 3 strategies for 2 days:
 - **Strategy 1:** 1 day in Bigtown, 1 day in Megalopolis.
 - **Strategy 2:** 2 days in Bigtown.
 - **Strategy 3:** 2 days in Megalopolis.
- They choose secretly at the same time, and the table shows how many votes (in thousands) Politician 1 wins (positive numbers) or loses (negative numbers) to Politician 2.

Understanding the Payoff Table

Politician 1 picks Strategy 1, Politician 2 picks Strategy 1: Politician 1 loses 3,000 votes (-3).

Politician 1 picks Strategy 2, Politician 2 picks Strategy 3: Politician 1 wins 2,000 votes (2).

Politician 1 picks Strategy 3, Politician 2 picks Strategy 3: Politician 1 loses 4,000 votes (-4).

This is a zero-sum game: Politician 2's gain is Politician 1's loss (e.g., if Politician 1 loses 3,000, Politician 2 wins 3,000).

■ TABLE 15.4 Payoff table for player 1 for variation 2 of the political campaign problem

Strategy	Player 2			Minimum
	1	2	3	
1	-3	-2	6	-3
2	2	0	2	0 ← Maximin value
3	5	-2	-4	-4
Maximum:	5	0	6	
		↑ Minimax value		

Breaking down the Numbers

- **Politician 1's Strategy 1:**
- **vs Politician 2**
 - Strategy 1: -3 (lose 3,000)
 - Strategy 2: -2 (lose 2,000)
 - Strategy 3: 6 (win 6,000)
- **Politician 1's Strategy 2:**
- **vs Politician 2**
 - Strategy 1: 2 (win 2,000)
 - Strategy 2: 0 (no change)
 - Strategy 3: 2 (win 2,000)

- **Politician 1's Strategy 3:**
- **vs Politician 2**
 - Strategy 1: 5 (win 5,000)
 - Strategy 2: -2 (lose 2,000)
 - Strategy 3: -4 (lose 4,000)

■ TABLE 15.4 Payoff table for player 1 for variation 2 of the political campaign problem

Strategy	Player 2			Minimum
	1	2	3	
1	-3	-2	6	-3
2	2	0	2	0 ← Maximin value
3	5	-2	-4	-4
Maximum:	5	0	6	
		↑ Minimax value		

Why No Dominated Strategies?

- In Variation 1, we dropped strategies that were always worse. Here:
 - Strategy 1 for Politician 1: Can win big (6) but risks losing (-3).
 - Strategy 2: Steady (never loses, max win is 2).
 - Strategy 3: High wins (5) but big losses (-4).
- No single strategy is always worse than another—it depends on what Politician 2 does. So, we can't just drop anything yet.

■ TABLE 15.4 Payoff table for player 1 for variation 2 of the political campaign problem

Strategy	Player 2			Minimum
	1	2	3	
1	-3	-2	6	-3
2	2	0	2	0 ← Maximin value
3	5	-2	-4	-4
Maximum:	5	0	6	
		↑ Minimax value		

Introducing the Minimax Criterion

- What is it?:
- A way to pick a strategy that's safe against a smart opponent.
 - Politician 1 **wants to maximize his minimum gain (avoid the worst loss).**
 - Politician 2 wants to **minimize his maximum loss (keep Politician 1's wins low).**
- Assumption: Both players are rational—they won't let the other win big if they can do it.

Politician 1's Thinking (Maximin)

- Look at the worst outcome for each strategy (minimum in each row):
 - Strategy 1: -3 (could lose 3,000 if Politician 2 picks Strategy 1).
 - Strategy 2: 0 (worst is breaking even if Politician 2 picks Strategy 2).
 - Strategy 3: -4 (could lose 4,000 if Politician 2 picks Strategy 3).
- Best of the worst: 0 (Strategy 2)—Politician 1 picks this because it guarantees no loss, even if Politician 2 is smart.

- Politician 1 wants to maximize his minimum gain (avoid the worst loss).
- Politician 2 wants to minimize his maximum loss (keep Politician 1's wins low).

■ TABLE 15.4 Payoff table for player 1 for variation 2 of the political campaign problem

		Player 2			Minimum
		1	2	3	
Player 1	1	-3	-2	6	-3
	2	2	0	2	0 ← Maximin value
	3	5	-2	-4	-4
Maximum:		5	0	6	
			↑		Minimax value

Politician 2's Thinking (Minimax)

- Look at the best outcome for Politician 1 in each column (maximum loss for Politician 2):
 - Strategy 1: 5 (Politician 1 could win 5,000 if they pick Strategy 3).
 - Strategy 2: 0 (Politician 1 wins nothing if they pick Strategy 2).
 - Strategy 3: 6 (Politician 1 could win 6,000 if they pick Strategy 1).
- Lowest of the best: 0 (Strategy 2)—Politician 2 picks this to keep Politician 1's win as small as possible.

Politician 1 wants to maximize his minimum gain (avoid the worst loss).

Politician 2 wants to minimize his maximum loss (keep Politician 1's wins low).

■ TABLE 15.4 Payoff table for player 1 for variation 2 of the political campaign problem

		Player 2			Minimum
		1	2	3	
Player 1	1	-3	-2	6	-3
	2	2	0	2	0 ← Maximin value
	3	5	-2	-4	-4
Maximum:		5	0	6	
			↑		Minimax value

The Solution

- Both pick Strategy 2 (2 days in Bigtown).
- Result: 0 votes change hands (Politician 1's payoff is 0).
- This is a fair game because the value (0) means no one wins or loses overall.

■ TABLE 15.4 Payoff table for player 1 for variation 2 of the political campaign problem

		Player 2			Minimum
		1	2	3	
Player 1	1	-3	-2	6	-3
	2	2	0	2	0 ← Maximin value
	3	5	-2	-4	-4
Maximum:		5	0 ↑ Minimax value	6	

Why This Works: Saddle Point

Notice the “0” in the table (Strategy 2 vs. Strategy 2)

- It's the minimum in its row (0 is the smallest of 2, 0, 2).
- It's the maximum in its column (0 is the largest of -2, 0, -2).
- This spot is called a saddle point—a stable solution.

Stable means: Neither politician can do better by switching:

- If Politician 1 switches to Strategy 1, he lose -2 (worse than 0).
- If Politician 2 switches to Strategy 1, Politician 1 wins 2 (worse for Politician 2).

What it all mean to us

Politician 1 picks Strategy 2
because it's the safest—no
risk of losing votes.

Politician 2 picks Strategy 2
because it stops Politician 1
from winning big.

They're stuck in a
stalemate—both playing it
safe leads to no vote change.

Variation 3 (When simple methods fail, needing mixed strategies)

Variation 3

- Unlike Variation 2, this version lacks a saddle point, making it unstable with pure strategies, and hints at the need for a more advanced approach)

■ **TABLE 15.5** Payoff table for player 1 for variation 3 of the political campaign problem

		Player 2			Minimum
		1	2	3	
Player 1	1	0	-2	2	-2 ← Maximin value
	2	5	4	-3	-3
	3	2	3	-4	-4
Maximum:		5	4	2	
				↑	
				Minimax value	

Some Example to Understand the Table

- Politician 1 picks Strategy 1, Politician 2 picks Strategy 2: Politician 1 loses 2,000 votes (-2).
- Politician 1 picks Strategy 2, Politician 2 picks Strategy 1: Politician 1 wins 5,000 votes (5).
- Politician 1 picks Strategy 3, Politician 2 picks Strategy 3: Politician 1 loses 4,000 votes (-4).
- Zero-Sum: Politician 2's gain equals Politician 1's loss (e.g., if Politician 1 wins 5,000, Politician 2 loses 5,000).

■ TABLE 15.5 Payoff table for player 1 for variation 3 of the political campaign problem

Strategy	Player 2			Minimum
	1	2	3	
1	0	-2	2	-2 ← Maximin value
2	5	4	-3	-3
3	2	3	-4	-4
Maximum:	5	4	2	
			↑ Minimax value	

Breaking down the numbers

- **Politician 1's Strategy 1:**
 - vs. Strategy 1: 0 (no change)
 - vs. Strategy 2: -2 (lose 2,000)
 - vs. Strategy 3: 2 (win 2,000)
- **Politician 1's Strategy 2:**
 - vs. Strategy 1: 5 (win 5,000)
 - vs. Strategy 2: 4 (win 4,000)
 - vs. Strategy 3: -3 (lose 3,000)

- **Politician 1's Strategy 3:**
 - vs. Strategy 1: 2 (win 2,000)
 - vs. Strategy 2: 3 (win 3,000)
 - vs. Strategy 3: -4 (lose 4,000)

■ **TABLE 15.5** Payoff table for player 1 for variation 3 of the political campaign problem

		Player 2			Minimum
		1	2	3	
Player 1	1	0	-2	2	-2 ← Maximin value
	2	5	4	-3	-3
	3	2	3	-4	-4
Maximum:		5	4	2	
				↑	Minimax value

Trying the Minimax Criterion

■ TABLE 15.5 Payoff table for player 1 for variation 3 of the political campaign problem

Strategy	Player 2			Minimum
	1	2	3	
1	0	-2	2	-2 ← Maximin value
2	5	4	-3	-3
3	2	3	-4	-4
Maximum:	5	4	2	
			↑ Minimax value	

- **Politician 1 (Maximin):** Pick the strategy with the best worst-case outcome.
 - Strategy 1 minimum: -2 (lose 2,000 if Politician 2 picks Strategy 2).
 - Strategy 2 minimum: -3 (lose 3,000 if Politician 2 picks Strategy 3).
 - Strategy 3 minimum: -4 (lose 4,000 if Politician 2 picks Strategy 3).
 - **Best minimum: -2 (Strategy 1)—Politician 1 picks this to limit losses to 2,000.**
- **Politician 2 (Minimax):** Pick the strategy that keeps Politician 1's maximum win lowest.
 - Strategy 1 maximum: 5 (Politician 1 wins 5,000 with Strategy 2).
 - Strategy 2 maximum: 4 (Politician 1 wins 4,000 with Strategy 2).
 - Strategy 3 maximum: 2 (Politician 1 wins 2,000 with Strategy 1).
 - **Lowest maximum: 2 (Strategy 3)—Politician 2 picks this to cap Politician 1's win at 2,000.**

But its unstable. Why?

■ TABLE 15.5 Payoff table for player 1 for variation 3 of the political campaign problem

Strategy	Player 2			Minimum
	1	2	3	
1	0	-2	2	-2 ← Maximin value
2	5	4	-3	-3
3	2	3	-4	-4
Maximum:	5	4	2	
			↑ Minimax value	

Politician 2's Reaction:

- He lose 2,000 votes and think, "I can do better!"
- Switch to Strategy 2: Now Politician 1 loses 2,000 (-2) instead of winning 2,000 – a 4,000-vote swing for Politician 2 (win 2,000 instead of lose 2,000).

Politician 1's Reaction:

- Sees the switch, switches to Strategy 2: Now wins 4,000 (4) instead of losing 2,000—a 6,000-vote improvement!

Politician 2 Again:

- Switches back to Strategy 3: Politician 1 loses 3,000 (-3) instead of winning 4,000.

And so on:

- Politician 1 might go back to Strategy 1, and the cycle repeats.

No Saddle Point

A saddle point is where the minimum of a row equals the maximum of a column (like Variation 2's "0").

No cell is both a row minimum and column maximum—no saddle point.

Without a saddle point, pure strategies (picking just one) don't settle the game

Instability: If Politician 1 knows Politician 2's move, he can switch to win more (e.g., from Strategy 1 to 2 against Strategy 2). Politician 2 can do the same. No single choice sticks.

What Next?

- This game can't be solved with one strategy—it's too unpredictable.
- The solution is to use **Mixed strategies**
- Instead of **picking one strategy, they mix them randomly** (e.g., 50% Strategy 1, 50% Strategy 2) to keep the opponent guessing.
- **They key take away point is:**
 - Variation 3 shows a game where pure strategies fail—smart players keep changing moves.
 - It's like rock-paper-scissors: If you always pick rock, they'll pick paper. You need to mix it up!

15.3: Games with Mixed Strategies

Game with Mixed strategies

This Section explains how to handle two-person, zero-sum games when choosing just **one strategy (a "pure strategy")** doesn't work

Because there's no saddle point – like in Variation 3 of the political campaign problem or the odds and evens game.

It introduces mixed strategies, where players randomly choose between their options using probabilities to keep their opponent guessing.

This section sets the stage for finding the best mix of strategies.

The Problem with Pure Strategies

- In games like Variation 3 (Table 15.5) or odds and evens (Table 15.1), picking one strategy (e.g., always showing 1 finger) fails:
 - **Odds and Evens Example:** If Player 1 always shows 1 finger, Player 2 can always show 2 fingers and win every time ($1+2 = 3$, odd, Player 2 wins).
 - **Variation 3 Example:** If Politician 1 picks Strategy 1, Politician 2 switches to Strategy 2 to win; if Politician 1 adjusts, Politician 2 switches again—no stable outcome.
- **Why?:**
 - No saddle point (where row minimum = column maximum), so pure strategies are predictable, and the opponent can exploit them.

What are Mixed Strategies

- Instead of picking one strategy every time, **players mix their choices** with probabilities.
- **What does it mean?**
 - Instead of choosing one pure strategy (action), players has randomized between strategies using certain probabilities
 - Because in some games, if you are predictable, your opponent can exploit you.
- By randomizing:
 - You become unpredictable and therefore your opponent cannot easily counter your move.
- **How It Works:**
 - Think of penalty kicks in football. The kicker can shoot left or right. If the kicker always shoots left, the goalkeeper will learn and always dive left.

What are Mixed Strategies (Cont..)

- **Definition of mix strategies:**

- Player 1 has strategies 1, 2, ..., m.
 - Assign probabilities $x_1, x_2, \dots, x_m$ (e.g., 50% Strategy 1, 50% Strategy 2) – be non negative.
- Player 2 has strategies 1, 2, ..., n. Assign probabilities $y_1, y_2, \dots, y_n$.
 - These probabilities must add to 1 (e.g., $x_1 + x_2 + \dots + x_m = 1$) and be non-negative.

- **How It Works:**

- Each player uses a random method (like flipping a coin) to pick a strategy based on these probabilities each time they play.

Example Mix Strategy in Variation 3

- Suppose Player 1: $(x_1, x_2, x_3) = (1/2, 1/2, 0)$
- (50% Strategy 1, 50% Strategy 2, 0% Strategy 3).
- Player 2: $(y_1, y_2, y_3) = (0, 1/2, 1/2)$
- (0% Strategy 1, 50% Strategy 2, 50% Strategy 3).
- Playing: Flip a coin—heads for Strategy 1, tails for Strategy 2 (Player 1); same for player 2 between Strategies 2 and 3.

■ **TABLE 15.5** Payoff table for player 1 for variation 3 of the political campaign problem

		Player 2			Minimum
		1	2	3	
Player 1	1	0	-2	2	-2 ← Maximin value
	2	5	4	-3	-3
	3	2	3	-4	-4
Maximum:		5	4	2	
				↑ Minimax value	

Why Mix Strategies?

- **Following reasons:**
 - **Unpredictability:** If your opponent can't guess your move, they can't always pick the perfect counter.
 - **Odds and Evens:** Always showing 1 finger loses predictably; flipping a coin (50% 1, 50% 2) keeps Player 2 from winning every time.
 - **Variation 3:** Mixing stops the endless switching – neither player can exploit a fixed choice.

Measuring Success: Expected Payoff

- The average votes Player 1 wins if the game is played many times with these probabilities.

$$\text{Expected payoff for player 1} = \sum_{i=1}^m \sum_{j=1}^n p_{ij} x_i y_j,$$

- Expected *payoff* = $p_{11}x_1y_1 + p_{12}x_1y_2 + p_{21}x_2y_1 + p_{22}x_2y_2$
- Where p_{ij} = the value in table representing the payoff with strategy i and j. For example, if player 1 chooses strategy 1 and player 2 strategy 2. they payoff is (-2) and so on
- x_i and y_j are the probabilities of each strategies of the player
- So we multiply payoff with the probabilities

$$\rightarrow (-2) (0.5) (0.5) + (2) (0.5) (0.5) + (4) (0.5) (0.5) + (-3) (0.5) (0.5)$$

$$0.250 = 250 \text{ votes}$$

■ **TABLE 15.5** Payoff table for player 1 for variation 3 of the political campaign problem

		Player 2			Minimum
		1	2	3	
Player 1	1	0	-2	2	-2 ← Maximin value
	2	5	4	-3	-3
	3	2	3	-4	-4
Maximum:		5	4	2	↑ Minimax value

Measuring Success: A simpler approach

- Example (Variation 3 Sample Mix):
- Each happens 25% of the time as $(1/2 \times 1/2 = 1/4)$ or $0.5 \times 0.5 = 0.25$
- Expected Payoff:
 $1/4(-2) + 1/4(2) + 1/4(4) + 1/4(-3)$
- ➔ $-0.5 + 0.5 + 1 - 0.75 = 0.25$ (250 votes on average).
- Note: This is just a sample mix—not the best yet.

$$\text{Expected payoff for player 1} = \sum_{i=1}^m \sum_{j=1}^n p_{ij} x_i y_j,$$

■ TABLE 15.5 Payoff table for player 1 for variation 3 of the political campaign problem

		Player 2			Minimum
		1	2	3	
Player 1	1	0	-2	2	-2 ← Maximin value
	2	5	4	-3	-3
	3	2	3	-4	-4
Maximum:		5	4	2	
				↑	Minimax value

Finding the Best Mix: Minimax Criterion

- The goal is
 - Player 1: Maximize the minimum expected payoff
 - Player 2: Minimize the maximum expected payoff to Player 1
- **Maximin Value ($\underline{v}$):** The smallest expected payoff Player 1 can guarantee.
- **Minimax Value ($\bar{v}$):** The largest expected loss Player 2 can force on Player 1.
- **Stable Solution:** When $\underline{v} = \bar{v} = v$ (the game's value) – neither can do better by switching alone.

Minimax Theorem

- **Statement:** With mixed strategies, there's always a pair of mixes where $\underline{v} = \overline{v} = v$, making it stable.
- **Proof:** Comes in [Section 15.5](#) – don't worry about it yet!

Why Randomness Even for One Game?

Intuition: If played once, why not just pick one strategy? Because if it's predictable, the opponent wins big.

Example: In odds and evens, always picking 1 loses; mixing 50/50 balances it.

Key: Randomness ensures the expected payoff holds, even if played once, by avoiding exploitation.

How to Find the Best Mix?

- Section 15.3 doesn't solve it yet—it sets up the idea.
- **Next Steps:**
 - Section 15.4: Graphical method (if one player has 2 strategies).
 - Section 15.5: Linear programming (for bigger games).

Section 15.4: Graphical Solution Procedure

Graphical Method

- **The graphical method** find the best mixed strategies for a game when one player **has just two options left after eliminating weaker (dominated)** strategies.
- Building on the Section 15.3's idea of mixed strategies let's use a graph
 - to figure out the optimal probabilities for each player,
 - ensuring a stable solution.
- The example is Variation 3 of the political campaign problem.

When to use This Method

- This method works for games where
 - One player has only 2 pure strategies (after dropping dominated ones).
- Why Graphical?:
 - Plotting lines on a graph makes it easier to see the best mix visually, rather than using complex math right away.
- Example used:
 - Variation 3 of the political campaign problem.

Original Table 15.5 and elimination the Dominated Strategy

- Check Player 1 strategies
- Strategy 3 (2, 3, -4) vs. Strategy 2 (5, 4, -3):
 - Column 1: $2 < 5$
 - Column 2: $3 < 4$
 - Column 3: $-4 < -3$
- Strategy 2 beats Strategy 3 every time
 - Strategy 3 is dominated, so drop it.

■ TABLE 15.5 Payoff table for player 1 for variation 3 of the political campaign problem

		Player 2			Minimum
		1	2	3	
Player 1	1	0	-2	2	-2 ← Maximin value
	2	5	4	-3	-3
	3	2	3	-4	-4
Maximum:		5	4	2	
				↑	Minimax value

Reduced Table

■ **TABLE 15.6** Reduced payoff table for player 1 for variation 3 of the political campaign problem

			Player 2		
		Probability	y_1	y_2	y_3
		Pure Strategy	1	2	3
<i>Player 1</i>	x_1	1	0	-2	2
	$1 - x_1$	2	5	4	-3

Calculating the Expected Payoff for Player 1

- **What's Expected Payoff?:** The average votes Player 1 wins as changes his strategies against Player 2's choice.
- 1,0,0 says player 2 chooses strategy 1,
0, 1, 0 says player chooses strategy and so on.
- Expected payoff equations calculates: payoff value into its probability of that strategy.
- i.e. $0 \times x_1 + 5(1 - x_1)$

Player 2 choices	(y_1, y_2, y_3)	Expected Payoff
	(1, 0, 0)	$0x_1 + 5(1 - x_1) = 5 - 5x_1$
	(0, 1, 0)	$-2x_1 + 4(1 - x_1) = 4 - 6x_1$
	(0, 0, 1)	$2x_1 - 3(1 - x_1) = -3 + 5x_1$

Expected payoff of Player 1

■ **TABLE 15.6** Reduced payoff table for player 1 for variation 3 of the political campaign problem

		Player 2		
		y_1	y_2	y_3
Probability		1	2	3
Player 1	Pure Strategy	0	-2	2
	Probability	5	4	-3

Calculating Expected Payoff for Player 1

What's Expected Payoff?: The average votes Player 1 wins if he play his mix many times against Player 2's pure strategies.

Player 2 picks Strategy 1:

- $\text{Payoff of Player 1} = 0 \cdot x_1 + 5 \cdot (1 - x_1) = 5 - 5 x_1$

- **Player 2 picks Strategy 2:**

- $\text{Payoff of player 1} = -2 \cdot x_1 + 4 \cdot (1 - x_1) = 4 - 6 x_1$

- **Player 2 picks Strategy 3:**

- $\text{Payoff of player 1} = 2 \cdot x_1 + (-3) \cdot (1 - x_1) = 2 x_1 - 3 + 3 x_1 = -3 + 5 x_1$

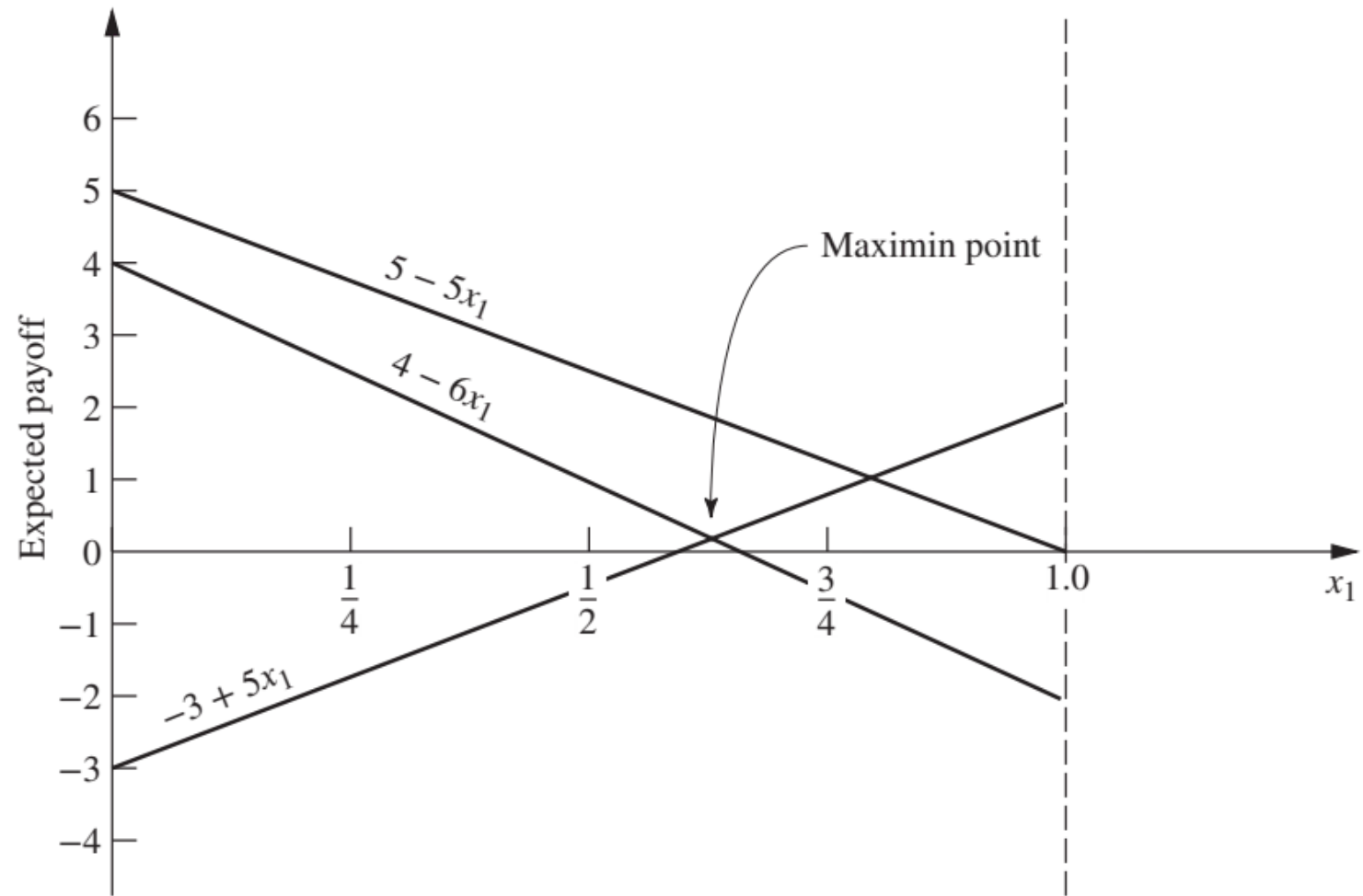
Plot the Line Graph

- **X-Axis:** x_1 (probability of Strategy 1, from 0 to 1).
- **Y-Axis:** Expected payoff for Player 1.
- **Three Lines** (one for each Player 2 strategy)
- Line 1 (Player 2 Strategy 1) $y = 5 - 5 x_1$
 - At $x_1 = 0$: $y = 5$
 - At $x_1 = 1$: $y = 0$

Plot the Line Graph (Cont...)

- **Line 2 (Player 2 Strategy 2):** $y = 4 - 6 x_1$
 - At $x_1 = 0$: $y = 4$
- At $x_1 = 1$: $y = -2$
- **Line 3 (Player 2 Strategy 3):** $y = -3 + 5 x_1$
 - At $x_1 = 0$: $y = -3$
 - At $x_1 = 1$: $y = 2$

Graph



■ **FIGURE 15.1**
Graphical procedure
for solving games.

Find Player 1's Best Mix (Maximin)

- **Goal:** Maximize the minimum expected payoff (best worst-case scenario).
- **Minimum Payoff:** For any x_1 , Player 2 picks the lowest line (since he wants Player 1 to win less).
 - Left side: Line 3 ($-3 + 5 x_1$) is lowest.
 - Right side: Line 2 ($4 - 6 x_1$) is lowest.
 - Line 1 ($5 - 5 x_1$) is never the lowest—it's always above the others.
- **Best Spot:** Where the two lowest lines (Line 3 and Line 2) meet – highest point.

Solving Best Maximin

- When the two lowest lines (Line 3 and Line 2) meet – is the highest point.
- At this point Line 2 and Line 3 are equal, therefore,
- $-3 + 5 x_1 = 4 - 6 x_1$
 - ➔ $5 x_1 + 6 x_1 = 4 + 3$
 - ➔ $11 x_1 = 7$
 - ➔ $x_1 = 7/11$ (about 0.636)
- Payoff at interaction
- $Y = -3 + 5 \cdot (7/11) \rightarrow -3 + 35/11 \rightarrow (-33 + 35)/11 \rightarrow 2/11 \rightarrow$ (about 0.182)
- Player 1 Mix : $x_1 = 7/11$, $1 - x_1 = 4/11$
- (Strategy 1 = 63.6% and Strategy 2 = 36.4%)

Player 2 Best Mix (Minimax)

- Taking the value of $x_1 = 7/11$ from the previous calculation, let's calculate the player 2 best mix.
- The goal is to minimize the maximum expected payoff of Player 1.
- Expected payoff is calculated by incorporating x_1 in the equation:

$$y_1 \cdot (5 - 5x_1) + y_2 \cdot (4 - 6x_1) + y_3 \cdot (-3 + 5x_1) \leq v = \frac{2}{11} \text{ for all } x_1$$

Evaluation: $y_1 \cdot \frac{20}{11} + y_2 \cdot \frac{2}{11} + y_3 \cdot \frac{2}{11}$

The key insight is that the sum of $y_1 + y_2 + y_3 = 1$

if $y_1 > 0$ the payoff exceeds $2/11$, therefore $y_1 = 0$

Solving the Equations

- $y_2 \cdot (4 - 6x_1) + y_3 \cdot (-3 + 5x_1) \leq v = \frac{2}{11}$ for all x_1
- At $x_1 = 0$: $4y_2 - 3y_3 = \frac{2}{11}$
- At $x_1 = 1$: $-2y_2 + 2y_3 = \frac{2}{11}$
- Also, $y_2 + y_3 = 1$

Solution:

- From $y_2 + y_3 = 1$ and $4y_2 - 3y_3 = 2/11$:
 - Solve: $4y_2 - 3(1 - y_2) = 2/11$
 - $4y_2 - 3 + 3y_2 = 2/11$
 - $7y_2 = 2/11 + 3 = 35/11$
 - $y_2 = 5/11, y_3 = 6/11$.

Player 2's Mix: $(y_1, y_2, y_3) = (0, 5/11, 6/11)$ (0% Strategy 1, 45.5% Strategy 2, 54.5% Strategy 3).

Result

- **Value of the Game:** $v = 2/11$ (about 182 votes on average for Player 1).
- **Stable:** Player 1's minimum payoff and Player 2's maximum loss both hit $2/11$ —no one can do better by switching alone (per the minimax theorem from Section 15.3).

How the graph Helps

- **Visual:** The graph shows three lines; (minimum payoff) peaks at $x_1 = 7/11$, $y = 2/11$
- **Player 2's Mix:** Weights Lines 2 and 3 to keep the payoff flat at $2/11$, ignoring Line 1 (which would raise it).

Key Takeaway

- The graphical method finds the best random mix when one player has 2 strategies.
- Player 1 mixes Strategies 1 and 2;
- Player 2 mixes Strategies 2 and 3 –
- This keeps the game fair and unpredictable.

Simple Summary

- **Section 15.4:** Use a graph to solve games with 2 strategies for one player.
- **Variation 3:** Drop Strategy 3, plot Player 1's payoffs, find $x_1 = 7/11$, $1 - x_1 = 4/11$ for Player 1, and $(0, 5/11, 6/11)$ for Player 2.
- **Result:** Player 1 wins $2/11$ (182 votes) on average—stable and smart!

Section 15.5. Solving by Linear Programming

Objectives of this section

- Learn how to use linear programming to solve two-person, zero-sum games.
- Understand the connection between game theory and linear programming.
- Here again we try to finding the Best Strategies
- Turn game theory into math you can solve!

Key Concepts

- **Key Concepts:**
 - Mixed Strategies
 - Minimax Theorem
 - Linear Programming Formulation

Why Use Linear Programming?

- In games **without a saddle point** (like Variation 3), a single "pure" strategy is unstable.
- Solving games with mixed strategies can become complex, especially for larger payoff matrices.
- **The Solution:** Players must use **Mixed Strategies** (probabilities over pure strategies).
- **The Problem:** How do we find the *optimal* probabilities for **large games** (more than 2 strategies)?
 - *Graphical Method* (Sec 15.4) only works when one player has 2 strategies.
 - **Solution:** Linear Programming works for *any* size game.

Advantages of Linear Programming:

- Provides a systematic and efficient way to find optimal mixed strategies.
- Can handle games with any number of strategies for both players.
- **Connection to Minimax Theorem:**
 - The minimax theorem guarantees that the optimal solution can be found using LP.

The Logic Behind the Transformation

We want to find the optimal strategy for **Player 1** ($x_1, \dots, x_m$) and the value of the game (v).

Key Insight:

If Player 1 uses an **optimal mixed strategy**, their expected payoff (v) must be **at least as good** as what they would get if Player 2 knew their plan and chose the *best response*.

- Mathematically: The expected payoff must be $\geq v$ **for every pure strategy** that Player 2 could play.

The Idea

- Mixed strategies = probabilities (e.g., x_1, x_2)
- Goal: Maximize Player 1's minimum payoff (v)
- Or minimize Player 2's maximum loss (v)
- Make sure v is the game's value—fair for both
- Balance scale with "Player 1" up and "Player 2" down

Deriving the Constraints

Let p_{ij} be the payoff if Player 1 uses strategy i and Player 2 uses strategy j .

If Player 2 chooses a **single pure strategy** j , the expected payoff for Player 1 is:

$$p_{1j}x_1 + p_{2j}x_2 + \cdots + p_{mj}x_m$$

To guarantee a minimum payoff of v against *any* move by Player 2, we need:

$$p_{1j}x_1 + p_{2j}x_2 + \cdots + p_{mj}x_m \geq v \text{ for } j = 1, 2, \dots, n$$

Plus the probability requirements:

$$x_1 + x_2 + \cdots + x_m = 1$$

$$x_i \geq 0 \text{ for all } i$$

The Linear Programming Problem (Player 1)

We don't know v , but we want to **maximize** it.

Let $x_{m+1} = v$.

Player 1's LP (Maximizing Player):

$$\text{Maximize } Z = x_{m+1}$$

Subject to:

$$p_{11}x_1 + p_{21}x_2 + \cdots + p_{m1}x_m - x_{m+1} \geq 0 \text{ (Col 1)}$$

$$p_{12}x_1 + p_{22}x_2 + \cdots + p_{m2}x_m - x_{m+1} \geq 0 \text{ (Col 2)}$$

$$\vdots$$

$$p_{1n}x_1 + p_{2n}x_2 + \cdots + p_{mn}x_m - x_{m+1} \geq 0 \text{ (Col } n)$$

$$x_1 + x_2 + \cdots + x_m = 1$$

$$x_i \geq 0, \text{ for } i = 1, \dots, m$$

(Note: x_{m+1} is unrestricted in sign, but we can adjust the table to make it non-negative).

The Dual Problem (Player 2)

Because this is a zero-sum game, Player 2's problem is the **Dual** of Player 1's problem.

Player 2's LP (Minimizing Player):

Let y_j be the probability Player 2 uses strategy j , and let $y_{n+1} = v$.

$$\text{Minimize } V = y_{n+1}$$

Subject to:

$$p_{11}y_1 + p_{12}y_2 + \cdots + p_{1n}y_n - y_{n+1} \leq 0 \text{ (Row 1)}$$

$$p_{21}y_1 + p_{22}y_2 + \cdots + p_{2n}y_n - y_{n+1} \leq 0 \text{ (Row 2)}$$

$$\vdots$$

$$p_{m1}y_1 + p_{m2}y_2 + \cdots + p_{mn}y_n - y_{n+1} \leq 0 \text{ (Row } m\text{)}$$

$$y_1 + y_2 + \cdots + y_n = 1$$

$$y_j \geq 0, \text{ for } j = 1, \dots, n$$

The Power of Duality

Because Player 1's LP and Player 2's LP are **Duals**:

1.Efficiency: Solving *one* LP (e.g., Player 1's) automatically gives the solution to the other.

2.Proof of Minimax Theorem: By the **Strong Duality Property** of Linear Programming:

$$\text{Max } x_{m+1} = \text{Min } y_{n+1}$$

1. Therefore, $\underline{v}$ (Maximin) = $\bar{v}$ (Minimax). This proves the existence of a stable solution in mixed strategies.

How it works?

- p_{ij} = payoff from Player 1's strategy i vs. Player 2's j
- Solve one problem – get both answers **(dual magic!)**
- $x_{m+1} + y_{n+1} = v$ (minimax theorem proof)

Handling Negative Values

- If v might be negative, adjust the table
- Add a big number (e.g., 3) to all payoffs
- Solve, then subtract it from v at the end
- Keeps the math working smoothly

■ **TABLE 15.6** Reduced payoff table for player 1 for variation 3 of the political campaign problem

		Player 2		
		y_1	y_2	y_3
	Probability			
	Pure Strategy	1	2	3
Player 1	x_1	0	-2	2
	$1 - x_1$	5	4	-3

Example Variation 3. Table 15. 6

- Player 1 mixes x_1, x_2 (probabilities)
- Player 2 mixes y_1, y_2, y_3
- Let's solve it!

■ **TABLE 15.6** Reduced payoff table for player 1 for variation 3 of the political campaign problem

		Player 2		
		y_1	y_2	y_3
	Probability			
	Pure Strategy	1	2	3
<i>Player 1</i>	x_1	0	-2	2
	$1 - x_1$	5	4	-3

Building the LP for the Example (Player 1)

Let x_1, x_2 be probabilities for Player 1. Let $x_3 = v$.

Objective function: Maximize x_3

Constraints: subject to:

- **Against Player 2 Strategy 1:** $0x_1 + 5x_2 - x_3 \geq 0 \rightarrow 5x_2 - x_3 \geq 0$
- **Against Player 2 Strategy 2:** $-2x_1 + 4x_2 - x_3 \geq 0$
- **Against Player 2 Strategy 3:** $2x_1 - 3x_2 - x_3 \geq 0$
- **Probability Sum:** $x_1 + x_2 = 1$
- **Non-negativity:** $x_1 \geq 0, x_2 \geq 0$

Building the LP for the Example (Player 2)

- Exercise

Player 2 Model (Variation 3)

- Minimize y_4

Constraints:

- $-2y_2 + 2y_3 - y_4 \leq 0$
- $5y_1 + 4y_2 - 3y_3 - y_4 \leq 0$
- $y_1 + y_2 + y_3 = 1$
- $y_1, y_2, y_3 \geq 0$

Dual of Player 1's problem

Solving the Simplex

- Use simplex method on Player 1's model
- Result: $x_1 = 7/11$, $x_2 = 4/11$, $x_3 = 2/11$
- Dual gives: $y_1 = 0$, $y_2 = 5/11$, $y_3 = 6/11$
- We have calculated these values in previous section

Checking the Value

- $v = x_3 = y_4 = \frac{2}{11}$ (about 182 votes)
- Matches minimax theorem ($\underline{v} = \bar{v}$)
- Stable solution—best for both!
- Politician 1 wins 182 votes on average

What if v is negative

- Add 3 to all Table 15.6 entries
- New $v = 35/11$, then subtract 3 $\rightarrow 2/11$
- Strategies stay the same
- Adjusts for negative payoffs safely

Why it is good approach

- Solves any size game (not just 2 strategies)
- One solve gets both players' answers
- Ties to duality theory
- This approach is powerful and efficient!

Practice Questions

1. Consider the game (the political game) with the payoff table (given in the chapter).
 - a. Use the concept of dominated strategies to eliminate any dominated strategies for Player 2, and show the reduced payoff table.
 - b. Find out the saddle point
 - c. Suppose the saddle point does not exist, apply the graphical solution procedure from Player 1's perspective
 - d. or use linear programming method to find optimal mix strategies.

Thanks